

PHYS 211 – Computational Homework #1

Infinite and Finite Quantum Wells

Objective:

In this assignment, you will numerically investigate and compare the quantum behavior of particles confined in infinite and finite square wells. You will explore quantized energies, wave functions, probability densities, tunneling, and the role of boundary conditions in quantum mechanics.

You may use MATLAB, Mathematica, or Python. You can use ChatGPT if you need help with the coding, but please disclose it in your submission.

Part I – Infinite Square Well

Consider a one-dimensional infinite square well:

$$U(x) = 0 \text{ for } 0 < x < L$$

$$U(x) = \infty \text{ otherwise}$$

with $L = 1.0 \text{ nm}$.

The normalized wavefunctions are:

$$\psi_n(x) = \sqrt{2/L} \sin(n\pi x/L)$$

and the energy levels are:

$$E_n = n^2 \pi^2 \hbar^2 / (2mL^2)$$

Assume the particle is an electron.

Problem 1 – Wave Functions and Probability Densities (20 pts)

- Plot the wave functions $\psi_1(x)$, $\psi_2(x)$, and $\psi_3(x)$ on the same graph.
- Plot the probability densities $|\psi_1(x)|^2$, $|\psi_2(x)|^2$, and $|\psi_3(x)|^2$ on the same graph.
- Describe how the number of nodes changes with increasing quantum number n .

Problem 2 – Energy Quantization (15 pts)

- Calculate the first three allowed energy levels in eV.
- Determine the wavelength of the photon emitted during the transition $n = 3 \rightarrow n = 1$.
- Explain qualitatively why the energy levels become more widely separated as n increases.

Part II – Finite Square Well

Now consider a finite square well:

$$U(x) = -U_0 \text{ for } |x| < a$$

$$U(x) = 0 \text{ for } |x| > a$$

with:

$$U_0 = 10 \text{ eV}$$

$$a = 0.5 \text{ nm}$$

Problem 3 – Finite Well Bound States (35 pts)

Using a numerical solver (or root-finding method), determine the bound-state energies of the finite well.

You may use:

- finite-difference methods,
- matrix eigenvalue methods,
- or numerical root solving.

- Find the first two bound-state energies.
- Plot the corresponding wave functions.
- Plot the corresponding probability densities.
- Show clearly that the wave functions extend into the classically forbidden region outside the well.

Problem 4 – Infinite vs Finite Well Comparison (20 pts)

- Why are the energy levels of the finite well lower than those of the infinite well?
- Why does tunneling occur in the finite well but not in the infinite well?
- Compare the shapes of the wave functions near the boundaries.
- Which system more closely resembles real physical systems? Explain.

Problem 5 – Reflection (10 pts)

Write a short paragraph discussing:

- what quantization means physically,
- why boundary conditions are important,

Submission Requirements

- Your code
- All plots
- Numerical calculations
- Written explanations

Constants

- $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
- $\hbar = 1.055 \times 10^{-34} \text{ J}\cdot\text{s}$
- $m_e = 9.11 \times 10^{-31} \text{ kg}$
- $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$
- $c = 3.00 \times 10^8 \text{ m/s}$

Notes

Clearly label all graphs and include axes with units. Include short interpretations and comments in your code explaining major steps.